

Total no. of Pages: .....

Roll no. ....

Degree: B.Tech (ECAM)

Semester: Third semester

**MID-SEMESTER EXAMINATION, September, 2024**

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Course Code: EAEPC305

Duration: 1:30 Hours

Max. Marks: 20

**Note:** - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

| Q. No. | Question                                                                                                      | Marks | CO   |
|--------|---------------------------------------------------------------------------------------------------------------|-------|------|
| 1a     | Given octal numbers $X = (263)_8$ and $Y = (157)_8$ .<br>Find their binary and decimal values.                | 2     | 1    |
| 1b     | If $P = 574$ and $Q = 683$ , find the BCD sum of P and Q.                                                     | 2     | 1    |
| 2a     | Given that $x = C8\text{ H}$ and $y = 7D\text{ H}$ ,<br>evaluate $(x - y)$ using 2's complement.              | 2     | 1    |
| 2b     | $F(A, B, C, D) = \sum \text{min} (1, 6, 8, 10, 14) + d(0, 2, 9)$ ;<br>Minimize using K-map.                   | 2     | 1    |
| 3a     | Design a 2-bit comparator logic circuit                                                                       | 2     | 1, 2 |
| 3b     | Draw a block diagram of a single digit BCD adder                                                              | 2     | 2    |
| 4a     | Implement using a 3 to 1 Multiplexer.<br>$F(A, B, C, D) = \sum \text{min} (0, 1, 3, 4, 5, 8, 10, 12, 14, 15)$ | 2     | 2    |
| 4b     | Design a Full Adder using a De-multiplexer                                                                    | 2     | 2    |
| 5a     | Explain the following characteristics of TTL logic family: Propagation delay and (ii) Noise margin            | 2     | 2    |
| 5b     | Design a CMOS circuit to implement:<br>$F(w, x, y) = (w + x.y + w.y)'$                                        | 2     | 2    |